

ACE INSTITUTE OF MANAGEMENT



Pokhara University
Faculty of Management



Bachelor in Computer System and Information Technology (BCSIT)

Project Proposal for Project II

Group-02

Submitted by

Anuska Khatri

Niharika Rajbhandari

Pranavi Uprety

Submitted to:

Sanju Shrestha

Certificate Page – Supervisor Recommendation

This project entitled 'Expense Management System (Sikka)' is recommended for evaluation and partial fulfillment of the academic requirements.

Certificate Page – Internal Examiner Approval

Approved by the Internal Examiner after evaluation of the project report and viva voce.

Abstract

The Expense Management System (Sikka) is a web-based application developed using PHP, MySQL, HTML, CSS, and JavaScript. The system enables users to register, authenticate, manage personal financial records, categorize expenses, maintain budgets, and monitor financial activities through a dashboard interface. The project aims to simplify personal financial management through an accessible and secure platform.

Acknowledgement

The successful completion of this project would not have been possible without the guidance of the supervisor, faculty members, colleagues, and family members who provided support throughout the development process.

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CHAPTER 1: INTRODUCTION

Background of the Study:

Digital technology has significantly transformed financial management practices. Individuals often face difficulties in tracking daily income and expenses, resulting in poor budgeting decisions. The Expense Management System provides a centralized platform for recording, monitoring, and analyzing financial activities.

Statement of the Problem:

Traditional methods of expense tracking are inefficient and prone to errors. Existing solutions may be expensive or overly complex for students and young professionals.

Objectives:

- Develop a web-based expense tracking platform.
- Provide secure user authentication.
- Manage income, expenses, categories, and budgets.
- Improve financial awareness.

Scope:

Personal expense tracking, income management, budget monitoring, dashboard visualization, and user account management.

Limitations:

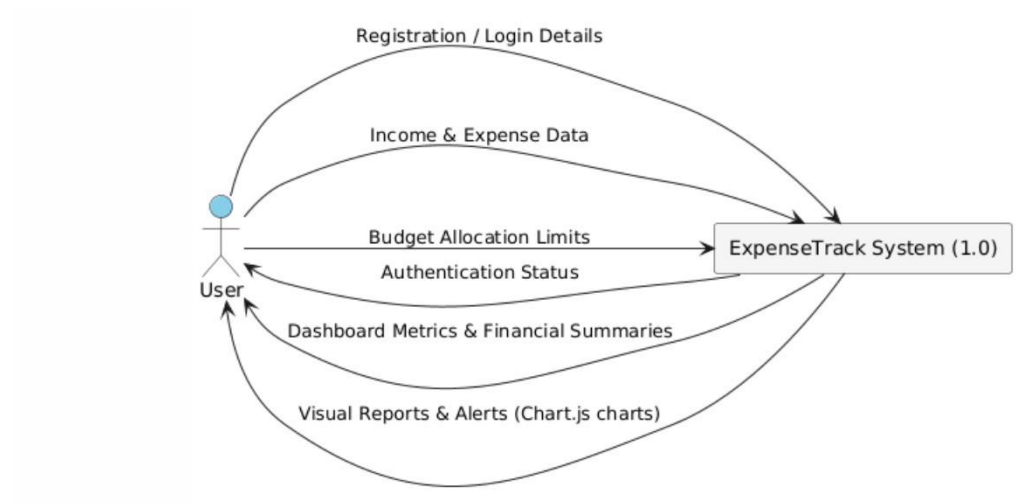
No bank integration, internet dependency, limited to registered users, and no mobile application version.

List of Abbreviations

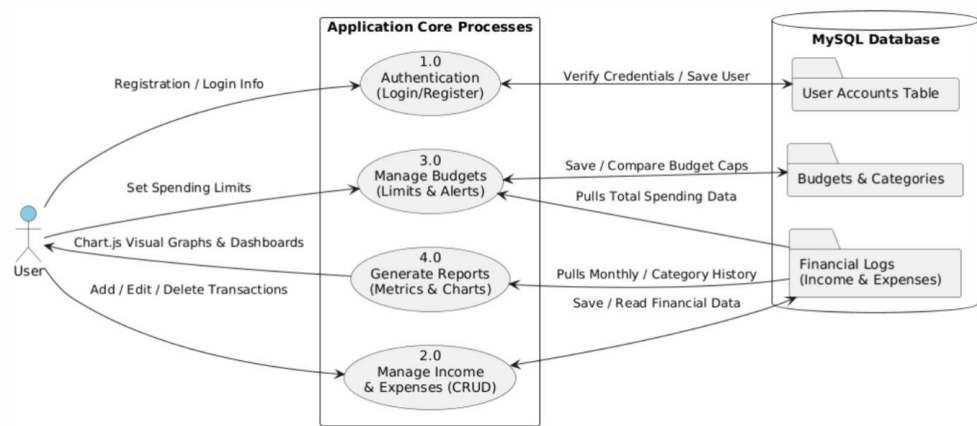
EMS	Expense Management System
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
PHP	Hypertext Preprocessor
SQL	Structured Query Language
MySQL	My Structured Query Language
DBMS	Database Management System
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram
UI	User Interface
UX	User Experience
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
IDE	Integrated Development Environment
VS Code	Visual Studio Code
XAMPP	Cross-Platform Apache MySQL PHP Perl

List of Figures

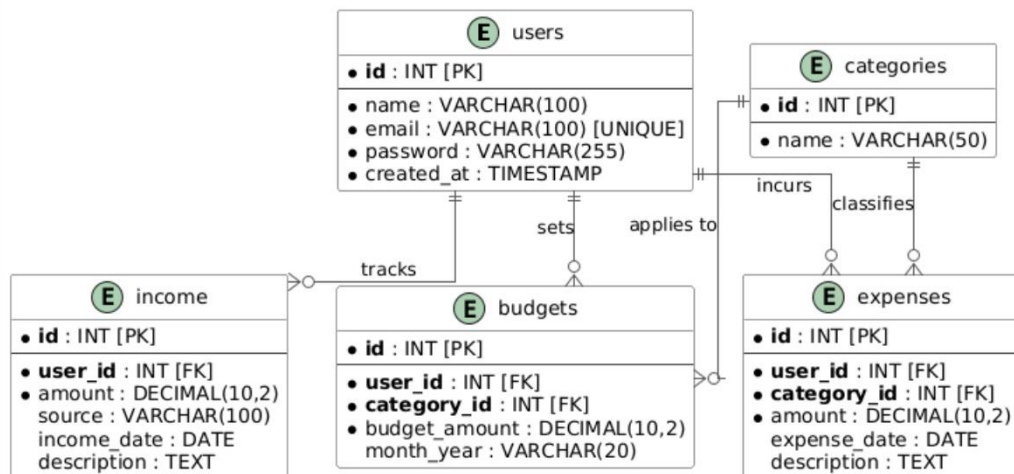
DFD level 0



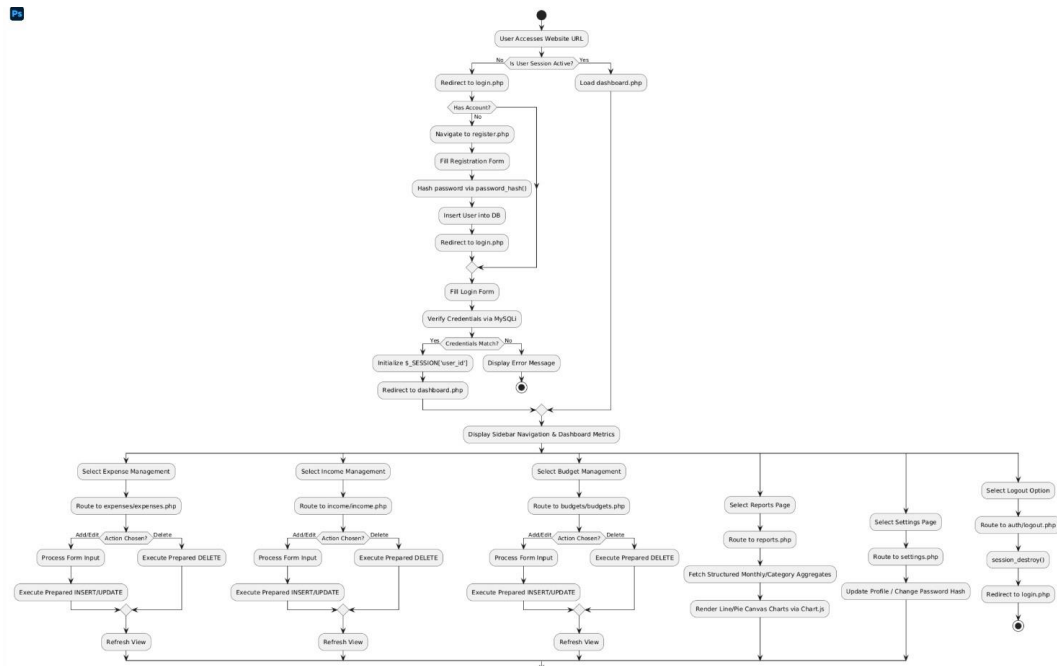
DFD level 1



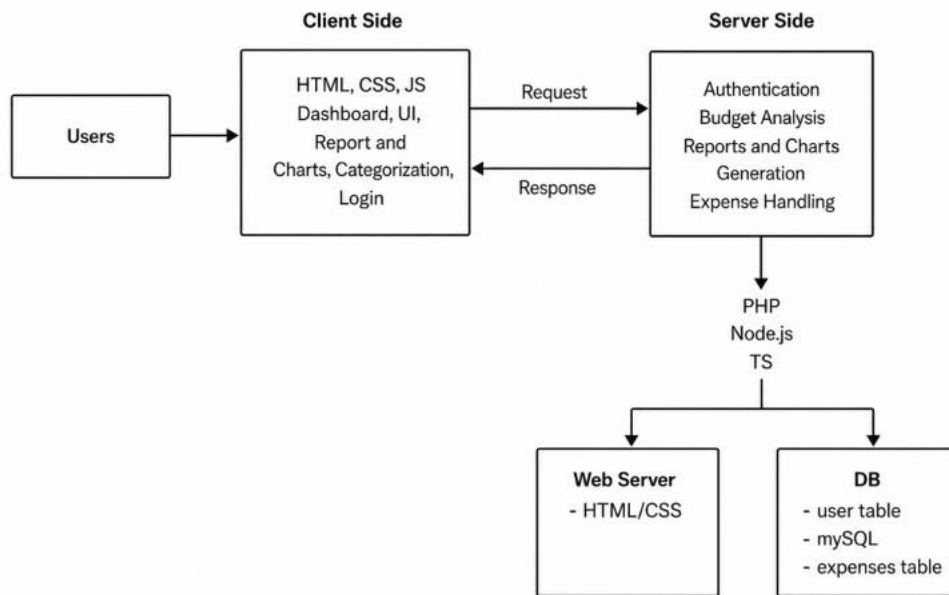
Use Case Diagram



Flowchart



System Architecture



CHAPTER 2: LITERATURE REVIEW

Expense management involves planning, recording, and analyzing financial transactions. Personal finance management systems help users monitor spending and improve savings habits.

Technologies:

HTML, CSS, JavaScript, PHP, MySQL.

Fundamental Concepts

- Expense Management
- Personal Finance Management
- Budgeting
- Group Expense Sharing
- Financial Reporting
- User Authentication
- Data Security

Review of Existing Systems

- A. Splitwise
- B. Goodbudget
- C. Spendee

Research Gap

Most existing systems focus either on personal budgeting or group expense management. The proposed system integrates both functionalities into a single platform while maintaining simplicity and affordability.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

System Analysis:

The proposed system replaces manual tracking methods with an automated database-driven solution.

Functional Requirements:

- User Registration
- User Login
- Dashboard View
- Income Management
- Expense Management
- Budget Management

Non-Functional Requirements:

Security, Reliability, Performance, Usability, Availability, and Scalability.

Feasibility Analysis:

Technical Feasibility: PHP and MySQL are readily available.

Operational Feasibility: Easy-to-use interface.

Financial Feasibility: Low development cost.

Schedule Feasibility: Achievable within academic project duration.

Data Modeling:

Entities include Users, Income, Expenses, Categories, and Budgets.

Database Schema:

Users

id (PK),name,email (UNIQUE),password (hashed)

created_at

Categories

id (PK),name,

Income

id (PK), user_id (FK → Users.id), amount, source, income_date

Expenses

id (PK), user_id (FK → Users.id), category_id (FK → Categories.id), amount, expense_date

Budgets

id (PK), user_id (FK → Users.id), category_id (FK → Categories.id), budget_amount

The screenshot shows a database management interface for a local server (localhost) connected to a database named 'expensetracker'. The 'Table: users' is selected, and the SQL query 'SELECT * FROM `users`' is entered. The interface displays a table with the following columns: id, name, email, password, and created_at. One row is visible, belonging to Anuska Khatri. The interface includes various tools like Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, Tracking, and Triggers. It also features a 'Query results operations' section with options like Print, Copy to clipboard, Export, Display chart, and Create view. At the bottom, there is a 'Bookmark this SQL query' section with a label input field and a checkbox to let every user access the bookmark.

id	name	email	password	created_at
1	Anuska Khatri	anushkakhatri25@gmail.com	\$2y\$10\$6ro6mNvPrghMnYDJAAuVn.Sw.DytjnlUN9r7MT7DeET...	2026-06-15 14:50:53

System Design:

Three-tier architecture consisting of Presentation Layer, Application Layer, and Database Layer.

List of Tables

Table No. Table Title

Table 3.1 Users Table Structure

Table 3.2 Income Table Structure

Table 3.3 Expenses Table Structure

Table 3.4 Categories Table Structure

Table 3.5 Budgets Table Structure

Table 4.1 Unit Testing Results

Table 4.2 System Testing Results

Table 4.3 Software and Hardware Requirements

CHAPTER 4: IMPLEMENTATION AND TESTING

Tools Used:

Frontend: HTML, CSS, JavaScript

Backend: PHP

Database: MySQL

Tools: VS Code, XAMPP, phpMyAdmin, Draw.io

Modules:

1. Registration Module
2. Login Module
3. Dashboard Module
4. Income Management Module
5. Expense Management Module
6. Budget Management Module

Testing:

Unit Testing:

TC01 Login Validation – Pass

TC02 Registration Validation – Pass

TC03 Database Connection – Pass

System Testing:

User registration, login, dashboard loading, transaction storage, and budget functionality were tested successfully.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

Lessons Learnt:

Requirement analysis, database design, web development, testing procedures, and project documentation.

Conclusion:

The Expense Management System successfully demonstrates a secure and user-friendly platform for personal financial management. It provides authentication, data management, and dashboard functionalities through a web-based environment.

Future Recommendations:

- Mobile Application Development
- AI-Based Expense Prediction
- Cloud Synchronization
- Multi-Currency Support
- Advanced Reporting Dashboard

REFERENCES

Splitwise. (2025). <https://www.splitwise.com>

Goodbudget. (2025). <https://goodbudget.com>

Spendee. (2025). <https://www.spendee.com>

APPENDICES

Appendix A: Screenshots

Appendix B: Source Code

Appendix C: Database Schema

Appendix D: Supervisor Visit Log Sheets